

Question 1: Long Answer Notes

Topics covered: (a) Blood - composition, functions, plasma and serum; (b) Blood pressure and long-term regulation; (c) Gastric juice, HCl secretion, gastric mucosal barrier and peptic ulcer.

How to use this note: For a 15-mark answer, start with definition, add labelled tables/flowcharts, write mechanism in steps, and end with applied importance. Underline normal values and keywords during revision.

1(a) Blood: composition, functions, plasma and serum

Definition and introduction

Blood is a fluid connective tissue circulating inside blood vessels. It is called the fluid of life because it transports oxygen from lungs to tissues and carbon dioxide from tissues to lungs. It is also called the fluid of growth because it carries absorbed nutrients and hormones to different parts of the body. It is called the fluid of health because it protects the body from infection and removes waste products through excretory organs.

In an average adult, blood volume is about 5 L and forms nearly 8% of body weight. Blood is slightly alkaline with a pH of about 7.4. It is more viscous than water because it contains red cells and plasma proteins.

Composition of blood

Blood has two major components: plasma and formed elements. Plasma is the straw-coloured liquid part, while formed elements include red blood cells, white blood cells and platelets.

Component	Approximate amount	Main contents
Plasma	About 55%	Water, plasma proteins, electrolytes, nutrients, hormones, gases and waste products
Formed elements	About 45%	RBCs, WBCs and platelets

Plasma

Plasma contains about 91-92% water and 8-9% solids. The solids are organic and inorganic substances. Organic substances include plasma proteins, amino acids, carbohydrates, lipids, enzymes, hormones, antibodies and non-protein nitrogenous substances such as urea, uric acid and creatinine. Inorganic substances include sodium, potassium, calcium, magnesium, chloride, bicarbonate, phosphate, iodide, iron and copper.

The major plasma proteins are albumin, globulin and fibrinogen. Albumin contributes most to plasma colloid osmotic pressure, globulins include antibodies and transport proteins, and fibrinogen is essential for clotting.

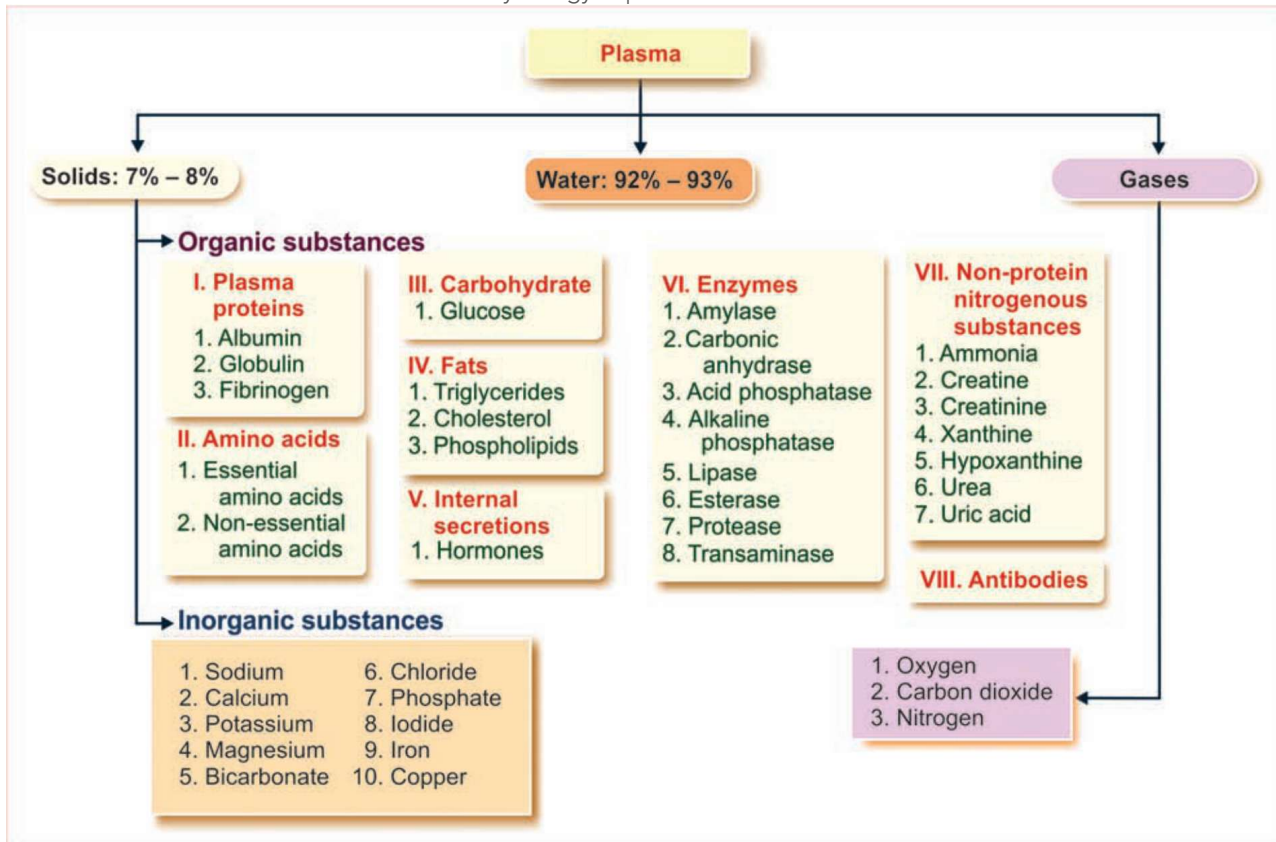


Figure: Composition of plasma from Sembulingam physiology source. Labels are kept large for easy reading.

Formed elements of blood

Element	Important feature	Main function
RBCs / Erythrocytes	Biconcave, non-nucleated cells containing haemoglobin	Transport oxygen and carbon dioxide
WBCs / Leukocytes	Nucleated defensive cells: neutrophils, eosinophils, basophils, lymphocytes and monocytes	Protection, phagocytosis, immunity and inflammation
Platelets / Thrombocytes	Small cell fragments derived from megakaryocytes	Haemostasis and blood clotting

Exam memory: Blood composition is commonly written as 55% plasma + 45% cells. Plasma = water + proteins + electrolytes + nutrients + gases + wastes.

Functions of blood

1. **Nutritive function:** Blood carries glucose, amino acids, fatty acids, glycerol, vitamins and minerals from the gastrointestinal tract to tissues for energy, growth and repair.
2. **Respiratory function:** Blood transports oxygen from lungs to tissues mainly through haemoglobin and transports carbon dioxide from tissues to lungs mainly as bicarbonate.
3. **Excretory function:** Blood carries urea, uric acid, creatinine, excess water and other metabolic wastes to kidneys, skin, liver and lungs for excretion.
4. **Transport of hormones and enzymes:** Hormones from endocrine glands are transported through blood to target tissues; enzymes and other signalling molecules are also carried.
5. **Regulation of water balance:** Water moves freely between blood and interstitial fluid, helping to maintain body fluid volume and tissue hydration.

6. Regulation of acid-base balance: Blood buffers such as bicarbonate, phosphate, haemoglobin and plasma proteins help maintain pH near 7.4.
7. Regulation of body temperature: Blood distributes heat from active organs such as liver and muscles to skin, where heat can be lost.
8. Storage function: Blood acts as a temporary reservoir for water and certain substances such as glucose and proteins.
9. Defensive function: WBCs, antibodies, complement and phagocytic cells protect the body against microorganisms and toxins.
10. Haemostatic function: Platelets and coagulation factors prevent blood loss after vessel injury by forming a clot.

Difference between plasma and serum

Point	Plasma	Serum
Definition	Liquid part of unclotted blood	Liquid part obtained after blood has clotted
Fibrinogen	Present	Absent
Clotting factors	Present	Mostly absent because they are consumed during clotting
Method of collection	Blood is mixed with anticoagulant and centrifuged	Blood is allowed to clot and then centrifuged
Use	Useful for coagulation studies and many blood tests	Useful for biochemical and serological tests
Formula	Plasma = serum + fibrinogen	Serum = plasma - fibrinogen

Serum is similar to plasma but lacks fibrinogen. This point is frequently asked as a short difference question and should be written in a table for quick marks.

1(b) Blood pressure: components and long-term regulation

Definition

Blood pressure is the lateral pressure exerted by flowing blood on the walls of blood vessels. In routine physiology, the term blood pressure usually refers to arterial blood pressure. Normal adult arterial blood pressure is about 120/80 mm Hg.

Components of blood pressure

Component	Meaning	Normal value	Key point
Systolic blood pressure	Maximum pressure during ventricular systole	About 120 mm Hg	Depends on force of contraction, stroke volume and arterial elasticity
Diastolic blood pressure	Minimum pressure during ventricular diastole	About 80 mm Hg	Depends mainly on peripheral resistance and heart rate
Pulse pressure	Difference between systolic and diastolic pressure	About 40 mm Hg	Pulse pressure = SBP - DBP
Mean arterial pressure	Average effective pressure in arteries during one cardiac cycle	About 93 mm Hg	MAP = DBP + 1/3 pulse pressure

Mean arterial pressure is more important for tissue perfusion than systolic or diastolic pressure alone because it represents the average driving pressure for blood flow throughout the cardiac cycle.

Factors determining arterial blood pressure

Arterial blood pressure depends mainly on cardiac output and total peripheral resistance. Cardiac output is affected by heart rate, stroke volume, venous return and blood volume. Peripheral resistance depends mainly on the diameter of arterioles and viscosity of blood.

Factor	Effect
Cardiac output	Directly proportional to blood pressure. Increased cardiac output raises BP.
Peripheral resistance	Increased arteriolar constriction increases diastolic pressure and MAP.
Blood volume	Increased blood volume increases venous return, cardiac output and BP.
Elasticity of arteries	Loss of elasticity increases systolic pressure and pulse pressure.
Viscosity of blood	Higher viscosity increases resistance and may raise BP.

Long-term regulation of arterial blood pressure

Long-term regulation of arterial blood pressure is mainly done by the kidneys. Nervous reflexes such as the baroreceptor reflex act rapidly, but their sensitivity decreases if pressure remains altered for a long time. Hence, renal mechanisms are essential for regulation over days, months and years.

The kidneys regulate arterial blood pressure by two major mechanisms: regulation of extracellular fluid volume and the renin-angiotensin-aldosterone mechanism.

1. Regulation of extracellular fluid volume

When arterial blood pressure increases, kidneys excrete more water and sodium. Increased water excretion is called pressure diuresis, while increased sodium excretion is called pressure natriuresis. Loss of water and sodium decreases extracellular fluid volume, blood volume, venous return and cardiac output. As a result, blood pressure returns towards normal.

When arterial blood pressure decreases, renal reabsorption of water and sodium increases. This increases ECF volume, blood volume, venous return and cardiac output, thereby restoring blood pressure. This renal-fluid mechanism is slow but very powerful.

Change	Renal response	Effect on volume	Final result
BP increases	More water and sodium excretion	ECF volume and blood volume decrease	BP falls toward normal
BP decreases	More water and sodium reabsorption	ECF volume and blood volume increase	BP rises toward normal

2. Renin-angiotensin-aldosterone mechanism

When blood pressure or ECF volume decreases, the juxtaglomerular apparatus of the kidney secretes renin. Renin converts angiotensinogen into angiotensin I. Angiotensin I is converted into angiotensin II by angiotensin-converting enzyme (ACE), mainly in the lungs and vascular endothelium.

Angiotensin II restores blood pressure in two important ways. First, it causes arteriolar vasoconstriction, increasing total peripheral resistance. Second, it stimulates the adrenal cortex to secrete aldosterone. Aldosterone increases sodium reabsorption from renal tubules; water follows sodium, increasing ECF volume and blood volume. This increases venous return, cardiac output and arterial blood pressure.

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renal tubules. Sodium reabsorption is followed by water reabsorption, resulting in increased

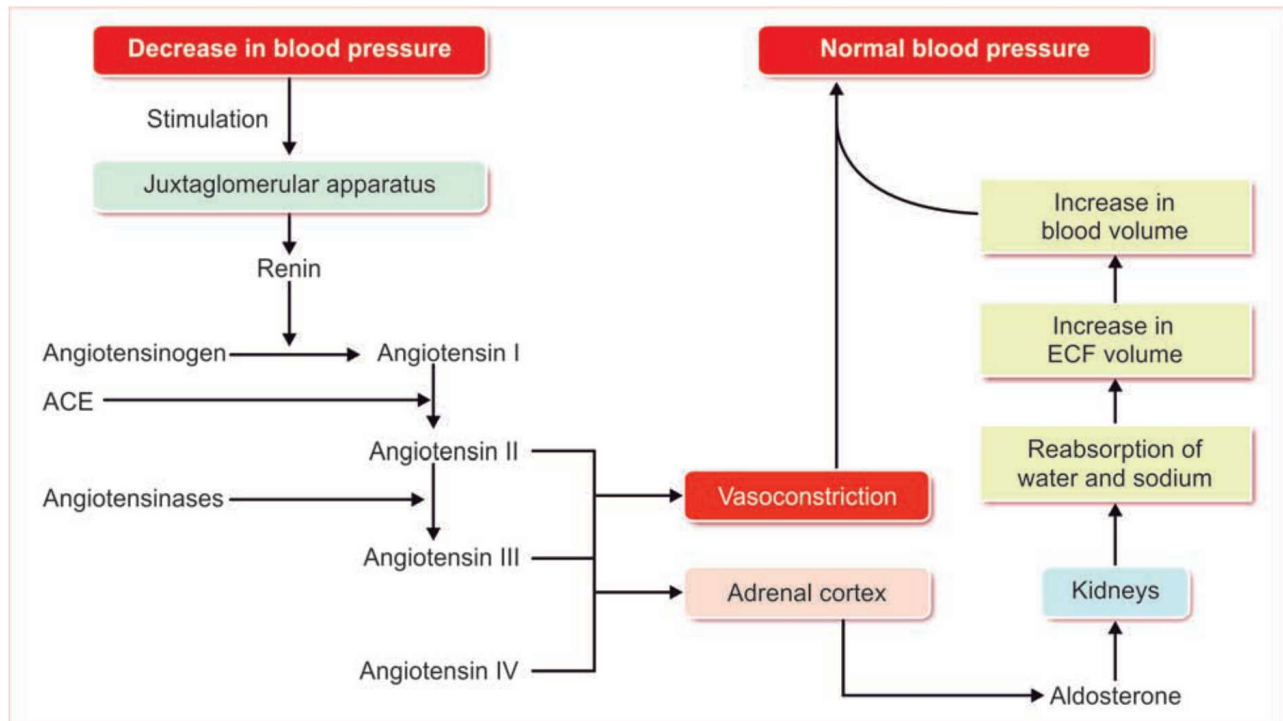


FIGURE 103.4: Regulation of blood pressure by renin-angiotensin mechanism. ACE = Angiotensin-converting enzyme.

Figure: Renin-angiotensin mechanism of long-term blood pressure regulation. Read the sequence from fall in BP to renin, angiotensin II, aldosterone, salt-water retention and restoration of BP.

Flowchart to write in exam

Decrease in BP or ECF volume → stimulation of juxtaglomerular apparatus → renin secretion → angiotensinogen to angiotensin I → ACE converts angiotensin I to angiotensin II → vasoconstriction + aldosterone secretion → sodium and water retention → increased blood volume and cardiac output → BP returns to normal.

Important distinction: Baroreceptor reflex is short-term. Renal-fluid mechanism and RAAS are long-term. In a long-answer question, give both renal mechanisms with a simple flowchart.

1(c) Gastric juice, HCl secretion, gastric mucosal barrier and peptic ulcer

Gastric juice: definition and properties

Gastric juice is the digestive secretion of the stomach. It is a mixture of secretions from different gastric glands present in the gastric mucosa. The daily volume is about 1200-1500 mL/day. It is highly acidic because of hydrochloric acid, with a pH around 0.9-1.2.

Constituents of gastric juice

Constituent	Function / significance
Water	Solvent for dissolved substances and enzymes
Hydrochloric acid	Secreted by parietal cells; activates pepsinogen and provides acidic medium
Pepsinogen	Secreted by chief cells; converted into pepsin for protein digestion
Gastric lipase	Weak lipolytic enzyme; digests some fats

Mucus / mucin	Protects gastric mucosa and lubricates food
Intrinsic factor of Castle	Secreted by parietal cells; needed for vitamin B12 absorption
Electrolytes	Na ⁺ , K ⁺ , Cl ⁻ , HCO ₃ ⁻ and other ions
Other enzymes	Gelatinase, urease, gastric amylase and rennin in infants/animals

Table: Secretory products of cells in gastric glands

Cell	Secretory products
Chief cells	Pepsinogen Rennin Lipase Gelatinase Urase
Parietal cells	Hydrochloric acid Intrinsic factor of Castle
Mucus neck cells	Mucin
G cells	Gastrin
Enterochromaffin (EC) cells	Serotonin
Enterochromaffin-like (ECL) cells	Histamine

Table: Secretory products of cells in gastric glands. Keep this ready for short notes and MCQs.

Functions of gastric juice

11. Digestive function: Pepsin digests proteins into proteoses, peptones and polypeptides. Gastric lipase helps in limited fat digestion.
12. Haemopoietic function: Intrinsic factor is required for absorption of vitamin B12 in the ileum. Deficiency may lead to pernicious anaemia.
13. Protective function: HCl kills many bacteria entering with food. Mucus and bicarbonate protect the gastric epithelium.
14. Activation of enzymes: HCl converts pepsinogen into active pepsin and provides the acidic pH required for pepsin action.
15. Excretory function: Some toxins, alkaloids and metals may be excreted through gastric juice.

Mechanism of HCl secretion

Hydrochloric acid is secreted by parietal cells of gastric glands. According to Davenport theory, it is an active process occurring in the canaliculi of parietal cells. Energy for the process is derived from cellular metabolism.

16. Carbon dioxide from metabolism or blood combines with water inside the parietal cell to form carbonic acid.
17. The reaction is catalysed by carbonic anhydrase, an enzyme present in high concentration in parietal cells.
18. Carbonic acid dissociates into hydrogen ions and bicarbonate ions.
19. Hydrogen ions are actively pumped into the canaliculus by the H⁺/K⁺ ATPase proton pump.
20. Chloride ions enter the parietal cell from blood and move into the canaliculus.
21. Inside the canaliculus, hydrogen ions combine with chloride ions to form HCl.

22. Bicarbonate ions enter the blood in exchange for chloride ions, producing a temporary rise in blood alkalinity after meals called alkaline tide.

Overall reaction: $\text{CO}_2 + \text{H}_2\text{O} + \text{NaCl} \rightarrow \text{HCl} + \text{NaHCO}_3$. This equation is useful in exams because it connects HCl formation with bicarbonate entry into blood.

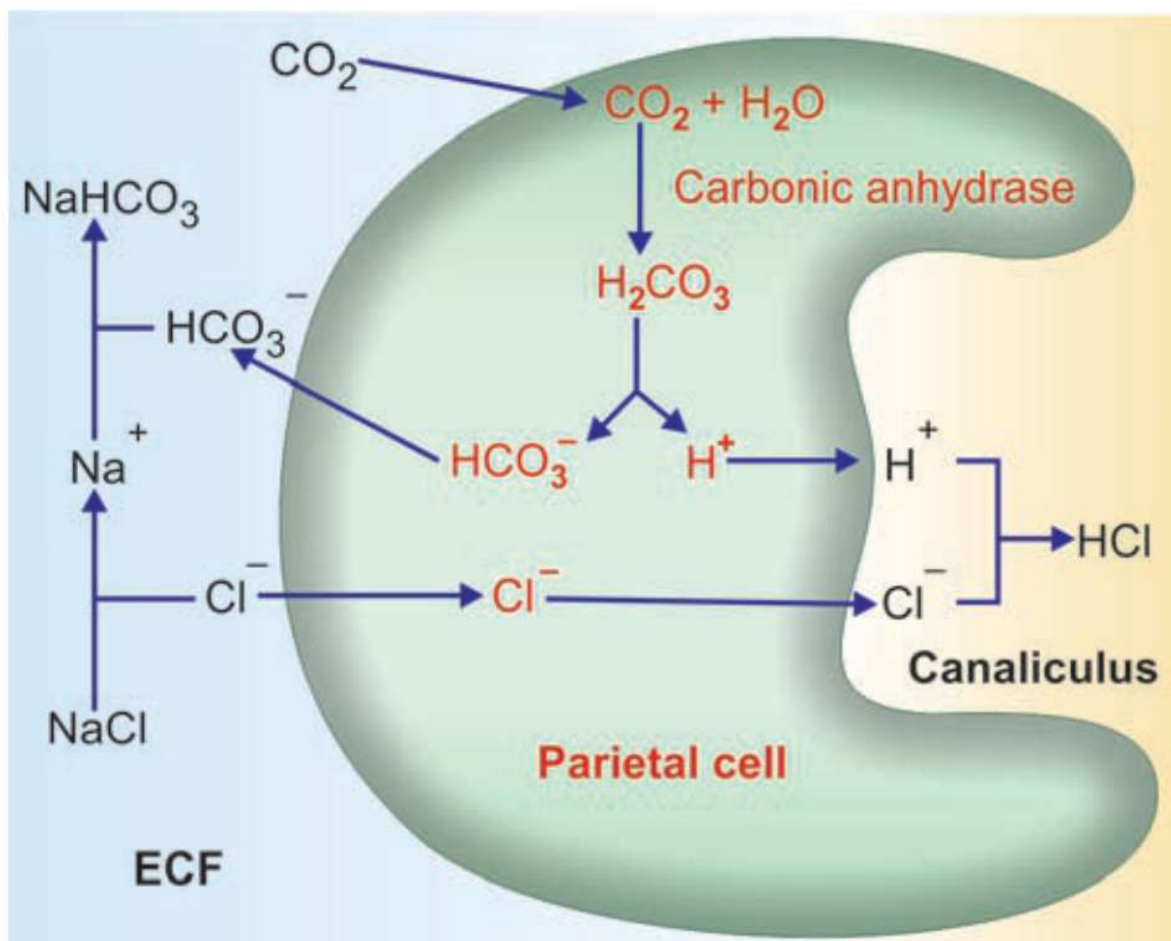


Figure: Secretion of hydrochloric acid by parietal cell. Important labels: carbonic anhydrase, H^+ , HCO_3^- , Cl^- , canaliculus and HCl.

Point	Details
Stimulators of HCl secretion	Gastrin, histamine and vagal stimulation
Inhibitors of HCl secretion	Secretin, gastric inhibitory polypeptide and peptide YY
Important drug target	H^+/K^+ ATPase proton pump is blocked by proton pump inhibitors

Gastric mucosal barrier

The gastric mucosal barrier is the protective mechanism that prevents the stomach from being damaged by its own acid and pepsin. The stomach normally secretes strong acid, but the mucosa remains protected because defensive factors form a barrier between acid and epithelial cells.

Component	Protective role
Mucus layer	Forms a thick protective coat over gastric mucosa and prevents direct contact of acid with epithelial cells
Bicarbonate secretion	Neutralizes acid trapped near the mucosal surface and maintains a relatively alkaline microenvironment
Tight junctions	Prevent back diffusion of hydrogen ions into deeper layers of mucosa
Rapid epithelial renewal	Damaged surface epithelial cells are quickly replaced
Adequate blood flow	Removes diffused acid and supplies oxygen and nutrients for repair
Prostaglandins	Increase mucus and bicarbonate secretion and help maintain mucosal blood flow

Clinically, NSAIDs may weaken the barrier because they reduce prostaglandin synthesis. This decreases mucus and bicarbonate secretion and increases the risk of gastritis and peptic ulcer.

Peptic ulcer

Peptic ulcer is an ulcer in a part of the gastrointestinal tract exposed to gastric acid and pepsin, commonly the stomach or the first part of the duodenum. It occurs when aggressive factors such as acid and pepsin overpower defensive factors such as mucus, bicarbonate, prostaglandins, epithelial repair and blood flow.

Aspect	Details
Aggressive factors	Hydrochloric acid, pepsin, Helicobacter pylori, NSAIDs, smoking, stress and alcohol
Defensive factors	Mucus, bicarbonate, epithelial barrier, prostaglandins, mucosal blood flow and rapid repair
Common sites	Stomach and duodenum
Main concept	Ulcer occurs when acid-pepsin injury is greater than mucosal protection

Final line for answer: Peptic ulcer is best explained as an imbalance: acid + pepsin + H. pylori/NSAIDs increase injury, while mucus + bicarbonate + prostaglandins + blood flow provide defence.

Last-minute revision sheet

Topic	Must-remember point
Blood composition	55% plasma, 45% formed elements; pH about 7.4; adult volume about 5 L
Plasma proteins	Albumin, globulin, fibrinogen
Plasma vs serum	Serum = plasma without fibrinogen/clotting factors
Normal BP	120/80 mm Hg
Pulse pressure	SBP - DBP = about 40 mm Hg
MAP	DBP + 1/3 pulse pressure = about 93 mm Hg
Long-term BP control	Kidneys: ECF volume regulation + RAAS
HCl source	Parietal cells
Pepsinogen source	Chief cells
Intrinsic factor source	Parietal cells
HCl pump	H ⁺ /K ⁺ ATPase
Mucosal barrier	Mucus + bicarbonate + tight junctions + blood flow + prostaglandins